

Gamma func. $\Gamma(x) = \int_0^\infty x^{-1} e^{-x} dx$
 1) $\Gamma(n) = (n-1)!$ $\exists \Gamma(1) = 1$
 2) $\Gamma(1/2) = \sqrt{\pi}$ $\Gamma(x) = (x-1)\Gamma(x-1)$
Distributions
 Uniform disc $\textcircled{D}$ Bernoulli $\textcircled{D}$ Binomial $\textcircled{D}$ Multinomial $\textcircled{D}$
 Normal $\textcircled{D}$ Poisson and exp $\textcircled{D}$ Chi-squared $\textcircled{D}$ Geometric $\textcircled{D}$

Uniform disc $\textcircled{D}$
 L r.v. assumes x_i w/ equal prob.
 L $f(x_i) = 1/k$, $x_i = x_1, x_2, \dots, x_k$
 L $\mu = \frac{1}{k} \sum_{i=1}^k x_i$
 L $\sigma^2 = \frac{1}{k} \sum_{i=1}^k (x_i - \mu)^2$
 L $M_x(t) = \frac{e^t(1 - e^{kt})}{k(1 - e^t)}$
Bernoulli $\textcircled{D}$
 L $x \in \{0, 1\}$
 L $f(x) = p^x (1-p)^{1-x}$
 L $\mu = p$
 L $\sigma^2 = p(1-p)$
 L $M_x(t) = 1 - p + pe^t$
Binomial $\textcircled{D}$
 L # of success in n indep. Bernoulli Trials
 L $b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}$, $x \leq n \in \mathbb{Z}^+$
 L $\mu = np$ L $\sigma^2 = npq$
 L $M_x(t) = (1 - p + pe^t)^n$
Multinomial $\textcircled{D}$
 L Trials w/ more than 2 possible outcomes
 L $f(x_1, x_2, \dots, x_k; p_1, p_2, \dots, p_k) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$
 L $\sum x_i = n$, $\sum p_i = 1$
Normal $\textcircled{D}$
 L Let X_i be indep. r.v. normal dist. w/ μ_i, σ_i^2
 L $Y = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2$ has chi-squared dist.
 L $v = n$ deg. of freedom
Poisson and exp $\textcircled{D}$
 1) $p(0; \lambda) = e^{-\lambda}$
 2) $c.d.f. = 1 - e^{-\lambda}$
 3) $p.d.f. = \lambda e^{-\lambda}$
Chi-squared $\textcircled{D}$
 L Prob. of x^{th} Bernoulli Trial w/ first success
 L $g(x; p) = b^*(x; p) = p q^{x-1}$, $x \geq 1 \in \mathbb{Z}^+$
 L $\mu = 1/p$, $\sigma^2 = (1-p)/p^2$
 L $M_x(t) = \frac{p}{1 - e^t + p e^t}$, $t < -\ln(1-p)$

Hypergeometric $\textcircled{D}$
 L Sampling w/o replacement (dep. trials)
 L $h(x; N, n, k) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}$
 L $\max\{0, n - (N - k)\} \leq x \leq \min\{n, k\}$
Poisson $\textcircled{D}$
 L Prob. of # of outcomes occurring in a given time interval / region t , for Poisson r.v. X
 L $\mu = \lambda t$
 L $M_x(s) = e^{\lambda t (e^s - 1)}$
 L $p(x; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$, $x = 0, 1, 2, \dots$
Chi-squared $\textcircled{C}$
 L $f(x; v) = \frac{1}{2^{v/2} \Gamma(v/2)} x^{v/2-1} e^{-x/2}$, $x > 0$
 L $\mu = v$, $\sigma^2 = 2v$
 L $M_x(s) = e^{ts - t^2/2}$
 L $\alpha = v/2$, $\beta = 2$
 L $v = \text{deg. of freedom}$
Uniform cont $\textcircled{C}$
 L $f(x; A, B) = \frac{1}{B-A}$, $A \leq x \leq B$
 L $\mu = (A+B)/2$ L $\sigma^2 = (B-A)^2/12$
 L $M_x(t) = (e^{Bt} - e^{At}) / t(B-A)$
Gamma $\textcircled{C}$
 L $f(x; \alpha, \beta) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}$, $x, \beta, \alpha > 0$
 L $\mu = \alpha\beta$ L $\sigma^2 = \alpha\beta^2$
 L $M_x(t) = \left(\frac{\beta}{1 - \beta t} \right)^\alpha$, $t < 1/\beta$
Exp. $\textcircled{C}$
 L $f(x; \beta) = \frac{1}{\beta} e^{-x/\beta}$, $x, \beta > 0$
 L $\mu = \beta$, $\sigma^2 = \beta^2$
 L $\alpha = 1$
Weibull $\textcircled{C}$
 L $f(x; \alpha, \beta) = \alpha \beta^\alpha x^{\alpha-1} e^{-\beta x^\alpha}$, $x, \beta, \alpha > 0$
 L $F(x) = 1 - e^{-\beta x^\alpha}$
 L $\mu = \alpha^{-1/\beta} \Gamma(1 + 1/\beta)$
 L $\sigma^2 = \alpha^{-2/\beta} [\Gamma(1 + 2/\beta) - \Gamma^2(1 + 1/\beta)]$
 L Failure rate $\lambda(t) = \alpha \beta t^{\alpha-1}$
 L $\beta = 1$, $\alpha > 1$, λ increasing, wear-in over t
 L $\beta > 1$, $\alpha < 1$, λ decreasing, wear-out over t
 L $\beta = 1$, $\alpha = 1$, λ constant, lack of memory

Normal $\textcircled{C}$
 L $\lambda(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{1}{2\sigma^2} (x - \mu)^2}$
 L Standardize: $z = (x - \mu) / \sigma$ or $x = z\sigma + \mu$
 L Percentile: $P(Z \leq z) = P(X \leq x) = \Phi$
Distribution properties
Hypergeometric, Binomial
 L If $n \ll N$ or $\frac{n}{N} \ll 0.05$
 L Then hypergeometric dist. $\approx$ binomial dist.
 L $p = k/N$
Binomial, Poisson
 L As $n \rightarrow \infty, p \rightarrow 0, np \rightarrow \mu$ (const)
 L $b(x; n, p) \approx p(x; \mu)$
 L $n \geq 20$ & $p \leq 0.05$ or $n \geq 100$ & $np \leq 10$
Standard Normal Binomial
 L Given X binomial, $\mu = np, \sigma^2 = npq$
 L As $n \rightarrow \infty$ (or $np \geq 5$ & $nq \geq 5$)
 L $P(a \leq X \leq b) \approx P\left(\frac{a - np - 0.5}{\sqrt{npq}} \leq Z \leq \frac{b - np + 0.5}{\sqrt{npq}}\right)$
Joint prob. distributions
Marginals
 L $f_X(x) = \sum_y f_{XY}(x, y)$ or $\int f_{XY}(x, y) dy$
 L $f_Y(y) = \sum_x f_{XY}(x, y)$ or $\int f_{XY}(x, y) dx$
Cond. dist.
 L $f(x|y) = f_{XY}(x, y) / f_Y(y)$
 L $f(y|x) = f_{XY}(x, y) / f_X(x)$
Statistical Mutual indep.
 L Given joint prob/pdft $f_{X_1, X_2, \dots, X_n}$
 L marginals $f_1(x_1), f_2(x_2), \dots, f_n(x_n)$
 L Indep. if $f_{X_1, X_2, \dots, X_n} = f_1(x_1) f_2(x_2) \dots f_n(x_n)$
 L $f(x|y) = f_X(x)$, vice versa
 L Joint dist $\rightarrow g(X, Y)$ & $h(X, Y)$
 L $E[g \pm h] = E[g] \pm E[h]$
 L $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$

Probability Theory
Cond. prob
 L $P(A|B) = \frac{P(A \cap B)}{P(B)}$
Mutual Exclusivity
 L $A \cap B = \emptyset$
 L $P(A \cup B) = P(A) + P(B)$
Mutual Independence
 L If $P(A \cap B) = P(A)P(B)$ or $P(A|B) = P(A)$, vice versa
Partition
 L If $A_1, A_2, \dots, A_n = S$
 L $\{A_i\}$ is a partition of S
 L Choosing
 L Comb, $nCr = \frac{n!}{(n-r)! r!}$
 L Perm, $nPr = \frac{n!}{(n-r)!}$
Set theory
 L $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 L $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$
Correlation coeff
 L $\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$ (no sq.)
 L 1) $\rho(aX + bY, cX + dY) = \rho(X, Y)$
 L 2) $-1 \leq \rho \leq 1$
Expectation
 L $\mu = E[X] = \int x f_X(x) dx$
 L 1) $E(aX + b) = aE[X] + b$
 L 2) $E(n) = n$
 L 3) $E[g(X) + h(X)] = \mu_g + \mu_h$
 L 4) For $X \geq 0$, $a, E(X) \geq 0$, $b, P(X=0) = 1$
Variance
 L $\sigma^2 = E[(X - \mu)^2] = E(X^2) - \mu^2$
 L 1) $\text{Var}(aX + b) = a^2 \sigma^2$
 L 2) $\text{Var}(X) \geq 0$
 L 3) $\text{Var}(X) = 0 \iff X = E(X), P(X=1)$
 L 4) $\text{Var}(n) = 0$
Covariance
 L $\sigma_{XY} = \text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X \mu_Y$
 L 1) $\text{Corr}(X, X) = \text{Var}(X)$
 L 2) $\text{Corr}(aX + b, cX + d) = ac \text{Corr}(X, Y)$
 L 3) $\text{Corr}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$
 L 4) Both large/small $\rightarrow$ +ve Cor.
 L One small one large $\rightarrow$ -ve Cor.
 L Indirect sum $\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$

Central Limit Theorem (One sample)

S-Mean $\bar{X}$; S-size n ; P-Mean μ ; P-var σ^2 ; $\sigma_x^2 = \frac{\sigma^2}{n}$
 As $n \rightarrow \infty$ (or $n \geq 30$), $Z = \frac{\bar{X} - \mu}{\sigma_x} \sim N(0,1)$
 Corollary: If $\bar{X} \sim N(\mu, \sigma_x^2)$ then $\sum (X_i - \mu)^2 \sim \chi^2(n, \sigma^2)$

Variance Dist

$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{\sigma^2}$
 $\chi^2 \sim \chi^2(n-1)$

Unbiased Estimator

$\hat{\theta}$ is an Unbiased Estimator for θ if $E[\hat{\theta}] = \theta$

Sample size (CI)

Given error e , $n = \frac{z_{\alpha/2}^2 \hat{p} \hat{q}}{e^2}$ exact
 or $n = \frac{z_{\alpha/2}^2}{4e^2}$ rounded up
 At least 100 (10%)

Central Limit Theorem (Two sample)

S-Mean $\bar{X}_1, \bar{X}_2$; S-size n_1, n_2 ; P-Mean μ_1, μ_2 ; P-var σ_1^2, σ_2^2 ; $\sigma_{\bar{X}_1 - \bar{X}_2}^2 = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$
 As $n \rightarrow \infty$ (or $n_1, n_2 \geq 30$), $Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}} \sim N(0,1)$

Corollary

① Rank data, $j=1$ lowest
 ② $f_i = (i-0.5)/n$ find z_i
 ③ x -axis: z_i , y -axis: actual value

Most efficient Estimator

For all unbiased estimators, the $\hat{\theta}$ w/ the smallest Var.

Maximum Likelihood Estimation

① Write Likelihood function $L(x_1, x_2, \dots, x_n; \theta) = f(x_1; \theta) f(x_2; \theta) \dots f(x_n; \theta)$
 ② Take Log
 ③ Differentiate find max

Law of Large #s

$\bar{X}_n \rightarrow E(\bar{X}_n) = \mu$ as $n \rightarrow \infty$
 $Var(\bar{X}_n) \rightarrow 0$

Chebyshev's Inequality

$P(\mu - k\sigma < \bar{X} < \mu + k\sigma) \geq (1 - \frac{1}{k^2})$ If X , non negative var and $E(X) < \infty, a > 0$

Markov's Inequality

Then $P(X \geq a) \leq E(X)/a$

Confidence Interval (CI)

$P(\hat{\theta}_L < \theta < \hat{\theta}_U) = 1 - \alpha$
 Confidence Limits: $\hat{\theta}_L, \hat{\theta}_U$
 Confidence Interval: $\hat{\theta}_U - \hat{\theta}_L$
 Confidence coeff: $1 - \alpha$
 ex. $f(x, p) = e^{-\lambda} \frac{\lambda^x}{x!}$ $\rightarrow L = \frac{e^{-\lambda} \lambda^x}{x!} \cdot \frac{e^{-\lambda} \lambda^x}{x!}$

Errors

	H_0 true	H_0 false	
Fail to reject H_0	✓	Type II error (miss)	$\alpha = P(\text{reject } H_0 H_0 \text{ true})$
Reject H_0	Type I error (false alarm)	✓	$\beta = P(\text{not reject } H_0 H_0 \text{ false})$ $= 1 - P(\text{reject } H_0 H_0 \text{ false})$ power

T-dist

If Z std normal, $V \sim \chi^2(v)$, $T = \frac{Z}{\sqrt{V/v}}$
 $L(t; v) = \frac{\Gamma((v+1)/2)}{\Gamma(v/2) \sqrt{v}} \left(1 + \frac{t^2}{v}\right)^{-(v+1)/2}$, $t \in \pm \infty$
 $L(t_1, -t_2) = -t_2$

F-dist

If $U \sim \chi^2(v_1)$, $V \sim \chi^2(v_2)$, $F = \frac{U/v_1}{V/v_2}$
 $L(f; v_1, v_2) = \begin{cases} \frac{\Gamma((v_1+v_2)/2)}{\Gamma(v_1/2) \Gamma(v_2/2)} \left(\frac{v_1}{v_2}\right)^{v_1/2} (1+f)^{-(v_1+v_2)/2} & f > 0 \\ 0 & f \leq 0 \end{cases}$
 $L(f_1, v_1, v_2) = [f_1(v_1, v_2)]^{-1} \rightarrow f^* = f(1, v)$

S-Mean and t-dist

$T = (\bar{X} - \mu) / (S/\sqrt{n}) \sim t(n-1)$

S-Var and F-dist

$F = \frac{\sigma_1^2 S_1^2 / q_1}{\sigma_2^2 S_2^2 / q_2} \sim F(q_1, q_2)$ Lowest level of significance

CI Table

Case C.I.
 σ known $\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

Pooled variance (For $\mu_1 = \mu_2$ CI)

$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1 + n_2 - 2}$

Sample Size (HT)

$\rightarrow$ One-sample or two for 2-tail
 $n \approx \left(\frac{z_{\alpha/2} + z_{\beta}}{\delta}\right)^2 \frac{\sigma^2}{\mu^2}$ $\delta = \mu_1 - \mu_0$

HT Table

Case	Statistic	H_0	H_1
One samp, known var	$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$	$\mu = \mu_0; -z_{\alpha/2} < z < z_{\alpha/2}$	$\mu \neq \mu_0; z > z_{\alpha/2} \text{ or } z < -z_{\alpha/2}$
One samp, unknown var	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, v=n-1$	$\mu = \mu_0; -t_{\alpha/2} < t < t_{\alpha/2}$	$\mu \neq \mu_0; t > t_{\alpha/2} \text{ or } t < -t_{\alpha/2}$
Two samp, known var	$z = \frac{\bar{x}_1 - \bar{x}_2 - \mu_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$	$\mu_1 = \mu_2; -z_{\alpha/2} < z < z_{\alpha/2}$	$\mu_1 \neq \mu_2; z > z_{\alpha/2} \text{ or } z < -z_{\alpha/2}$
Two samp, equal unknown var (pooled t-test)	$t = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{s_p \sqrt{1/n_1 + 1/n_2}}, v = n_1 + n_2 - 2$	$\mu_1 = \mu_2; -t_{\alpha/2} < t < t_{\alpha/2}$	$\mu_1 \neq \mu_2; t > t_{\alpha/2} \text{ or } t < -t_{\alpha/2}$
Two samp, unequal unknown var	$t = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}, v = \frac{n_1 n_2}{n_1 + n_2}$	$\mu_1 = \mu_2; -t_{\alpha/2} < t < t_{\alpha/2}$	$\mu_1 \neq \mu_2; t > t_{\alpha/2} \text{ or } t < -t_{\alpha/2}$
Paired obs	$t = \frac{\bar{d} - \mu_0}{s_d/\sqrt{n}}, v=n-1$	$\mu_0 = d_0; -t_{\alpha/2} < t < t_{\alpha/2}$	$\mu_0 \neq d_0; t > t_{\alpha/2} \text{ or } t < -t_{\alpha/2}$

σ unknown $\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}}, v=n-1$

Pred. int. σ known $\bar{x} - z_{\alpha/2} \sigma \sqrt{1/n + 1/n_0} < x_0 < \bar{x} + z_{\alpha/2} \sigma \sqrt{1/n + 1/n_0}$

Pred. int. σ unknown $\bar{x} - t_{\alpha/2} s \sqrt{1/n + 1/n_0} < x_0 < \bar{x} + t_{\alpha/2} s \sqrt{1/n + 1/n_0}, v=n-1$

$\mu_1 - \mu_2$ σ known $(\bar{x}_1 - \bar{x}_2) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

$\mu_1 - \mu_2$ σ unknown, eq $(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$

$\mu_1 - \mu_2$ σ unknown, uneq $(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

$v = \frac{(s_1^2/n_1 + s_2^2/n_2)}{[(s_1^2/n_1)/(n_1-1)] + [(s_2^2/n_2)/(n_2-1)]}$ rounded down

Variance σ^2 $(n-1)s^2/x_{\alpha/2}^2 < \sigma^2 < (n-1)s^2/x_{1-\alpha/2}^2, v=n-1$

Ratio of Variances σ_1^2/σ_2^2 $s_1^2/s_2^2 \cdot \frac{1}{F_{\alpha/2}} < \sigma_1^2/\sigma_2^2 < s_1^2/s_2^2 \cdot \frac{1}{F_{1-\alpha/2}}, v_1=n_1-1, v_2=n_2-1$

Proportion p $\hat{p} - z_{\alpha/2} \sqrt{\hat{p}\hat{q}/n} < p < \hat{p} + z_{\alpha/2} \sqrt{\hat{p}\hat{q}/n}, \hat{q} = 1 - \hat{p}$

Proportion $p_1 - p_2$ $(\hat{p}_1 - \hat{p}_2) - z_{\alpha/2} \sqrt{\hat{p}_1 \hat{q}_1/n_1 + \hat{p}_2 \hat{q}_2/n_2} < p_1 - p_2 < (\hat{p}_1 - \hat{p}_2) + z_{\alpha/2} \sqrt{\hat{p}_1 \hat{q}_1/n_1 + \hat{p}_2 \hat{q}_2/n_2}$

Paired obs $\bar{d} - t_{\alpha/2} \frac{s_d}{\sqrt{n}} < \mu_0 < \bar{d} + t_{\alpha/2} \frac{s_d}{\sqrt{n}}, \bar{d} = \sum d_i/n, s_d = \sqrt{\sum (d_i - \bar{d})^2 / (n-1)}$

β_1 $b_1 - t_{\alpha/2} \frac{s_{b_1}}{\sqrt{S_{xx}}} < \beta_1 < b_1 + t_{\alpha/2} \frac{s_{b_1}}{\sqrt{S_{xx}}}$ $\rightarrow$ std error, slope

β_0 $b_0 - t_{\alpha/2} \frac{s_{b_0}}{\sqrt{S_{xx}}} < \beta_0 < b_0 + t_{\alpha/2} \frac{s_{b_0}}{\sqrt{S_{xx}}}$ $\rightarrow$ std error intercept

Mean response $\hat{y}_0$ $\hat{y}_0 - t_{\alpha/2} s_{\hat{y}_0} < \mu_{\hat{y}_0} < \hat{y}_0 + t_{\alpha/2} s_{\hat{y}_0}, s_{\hat{y}_0}^2 = s^2 \left[\frac{1}{n} + \frac{(\bar{x}_0 - \bar{x})^2}{S_{xx}} \right], v=n-2$

Single response y_0 $\hat{y}_0 - t_{\alpha/2} s_{y_0} < y_0 < \hat{y}_0 + t_{\alpha/2} s_{y_0}, s_{y_0}^2 = s^2 \left[1 + \frac{(\bar{x}_0 - \bar{x})^2}{S_{xx}} \right], v=n-2$

Fit Line Formulas

$\hat{y} = b_0 + b_1 x$

$y_i = b_0 + b_1 x_i + e_i$ residual

$b_1 = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2} \sim (\beta_1, \frac{\sigma^2}{S_{xx}})$

$b_0 = \bar{y} - b_1 \bar{x} \sim (\beta_0, \sigma^2 \sum x_i^2 / n S_{xx})$

$s^2 = SSE/(n-2), E(s^2) = \sigma^2$ Mean Sq Error

$R^2 = 1 - SSE/SST$ Coeff. of determination

$= SSR/SST$ measured with y explained by x

$= 1 - \frac{s^2(n-2)}{SST}$

Sum of Squares (SS)

$S_{xx} = \sum (x_i - \bar{x})^2$

$S_{yy} = \sum (y_i - \bar{y})^2$

$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$

$SSE = \sum e_i^2 = S_{yy} - b_1 S_{xy}$ Error $v=n-2$

$SSR = \sum (\hat{y}_i - \bar{y})^2$ Regression $v=1$

$SST = \sum (y_i - \bar{y})^2 = SSE + SSR$ Total $v=n-1$

Sample Variance

$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$

Same as ① y related when $\mu_0 = 0$

Lack of fit $h = \text{rep obs.}$

Correlation test, ρ

Int, β_0

$f = \frac{SSR}{s^2}, v_1=1, v_2=n-2$

$f = \frac{SSE - PSSE}{s^2}, v_1=h-2, v_2=n-h$

$t = \frac{\hat{\beta}_0 - \mu_0}{s_{\hat{\beta}_0}}, v=n-2$

$\beta_0 = \beta_0; -t_{\alpha/2} < t < t_{\alpha/2}$

$\beta_0 \neq \beta_0; t > t_{\alpha/2} \text{ or } t < -t_{\alpha/2}$

$\beta_1 = 0; f \leq f_{\alpha}$

$\beta_1 \neq 0; f > f_{\alpha}$

$\beta_1 \neq 0; t > t_{\alpha/2} \text{ or } t < -t_{\alpha/2}$

$\rho = 0; t > t_{\alpha/2} \text{ or } t < -t_{\alpha/2}$

$\rho \neq 0; t > t_{\alpha/2} \text{ or } t < -t_{\alpha/2}$

Multiple Measurements

$n_i \rightarrow \# \text{ obs. for } x=x_i, \sum_{i=1}^h n_i = n$ (total)

$y_{ij} \rightarrow j \text{th value for } Y_i; y_i = \sum_{j=1}^{n_i} y_{ij} / n_i$

$PSSE = \sum_{i=1}^h \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2, v = n-h$

$s^2 = PSSE/n-h = \frac{\sum_{i=1}^h \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2}{(n-h)}$

$E(s^2) = \sigma^2$

Lack of fit $\rightarrow$ Lot $SS = SSE - PSSE = \sum_{i=1}^h n_i (\bar{y}_i - \bar{y})^2, v=h-2$

$\rho = h-2$

$\rho \neq h-2$

$\rho \neq h-2$

$\rho \neq h-2$

Lambda code

① $t = \frac{\text{estimate}}{\text{std. error}}$

② Adj $R^2 =$

③ $f = \text{Mean sq. Y} / \text{Mean sq. Res.}$

④ $SSE \rightarrow \text{Res. Sum Sq}$

⑤ $SSR \rightarrow y$ Sum Sq

⑥ Res. std. error = $\sqrt{\text{Res. Mean Sq}}$

⑦ Mean Sq. Error = $\frac{\text{Sum sq}}{\text{df}}$

⑧ $1 - \left[\frac{(1-R^2)(n+1)}{n-2} \right]$

$\rightarrow$ as n at indep var. x_1, x_2 etc.

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